IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>
<Date>

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies： Data was gathered via the SpaceX REST API and Wikipedia web scraping, wrangled into binary landing classes, analyzed using SQL (SQLite) and Python (Seaborn), mapped geospatially with Folium, dashboarded in Plotly Dash, and modeled using tuned classification algorithms (LR, SVM, Decision Tree, KNN).
- Summary of all results： KSC LC-39A led launchpads with a 76.9% success rate, geospatial mapping confirmed landing complexes are placed near coastlines to maximize safety, and all optimized machine learning models converged on an identical prediction accuracy of 83.33%.

Introduction

- Project background and context： SpaceX reduces launch costs to $62M (vs. $165M+ from competitors) by recovering and reusing the Falcon 9 first-stage booster. This capstone applies data science to predict booster reusability.
- Problems you want to find answers： How do flight variables (payload, orbit, launch site) impact landing success, can we build an ML classifier to predict booster recovery, and how can competitors use this to bid strategically?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Conducted REST API calls (GET requests) to the official SpaceX data endpoint to extract historical core data, flight details, payloads, launch blocks, and landing outcomes. Formulated a scraping pipeline using BeautifulSoup to extract additional Falcon 9 launch tables from Wikipedia, capturing missing structural parameters and early launch variables.
- Perform data wrangling
  - Handled null metrics across payload masses by replacing them with calculated mean values for their respective booster variants or orbits. Decoded string outcome parameters (e.g., *Success (drone ship)*, *Failure (parachute)*) into a standardized binary outcome classifier Class (1 for successful stage landings, 0 for unsuccessful or unattempted landings).

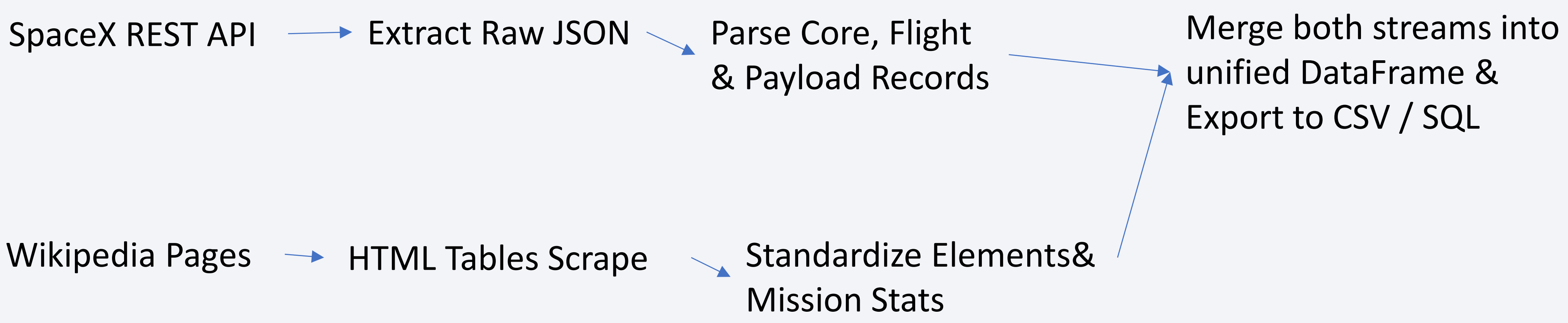
Methodology

Executive Summary

- Perform exploratory data analysis (EDA) using visualization and SQL
  - Produced scatter plots, Managed data verification inside an SQLite instance
- Perform interactive visual analytics using Folium and Plotly Dash
  - Rendered interactive maps tracking worldwide launch facilities, Built a web application dashboard with interactive dropdown filters and a sliding payload mass scale
- Perform predictive analysis using classification models
  - Constructed four distinct predictive machine learning classification pipelines, Wrapped each pipeline in a systematic GridSearchCV routine to fine-tune regularization parameters and tree depths, ultimately evaluating model quality using visual confusion matrices and cross-validated accuracy scores.

Data Collection

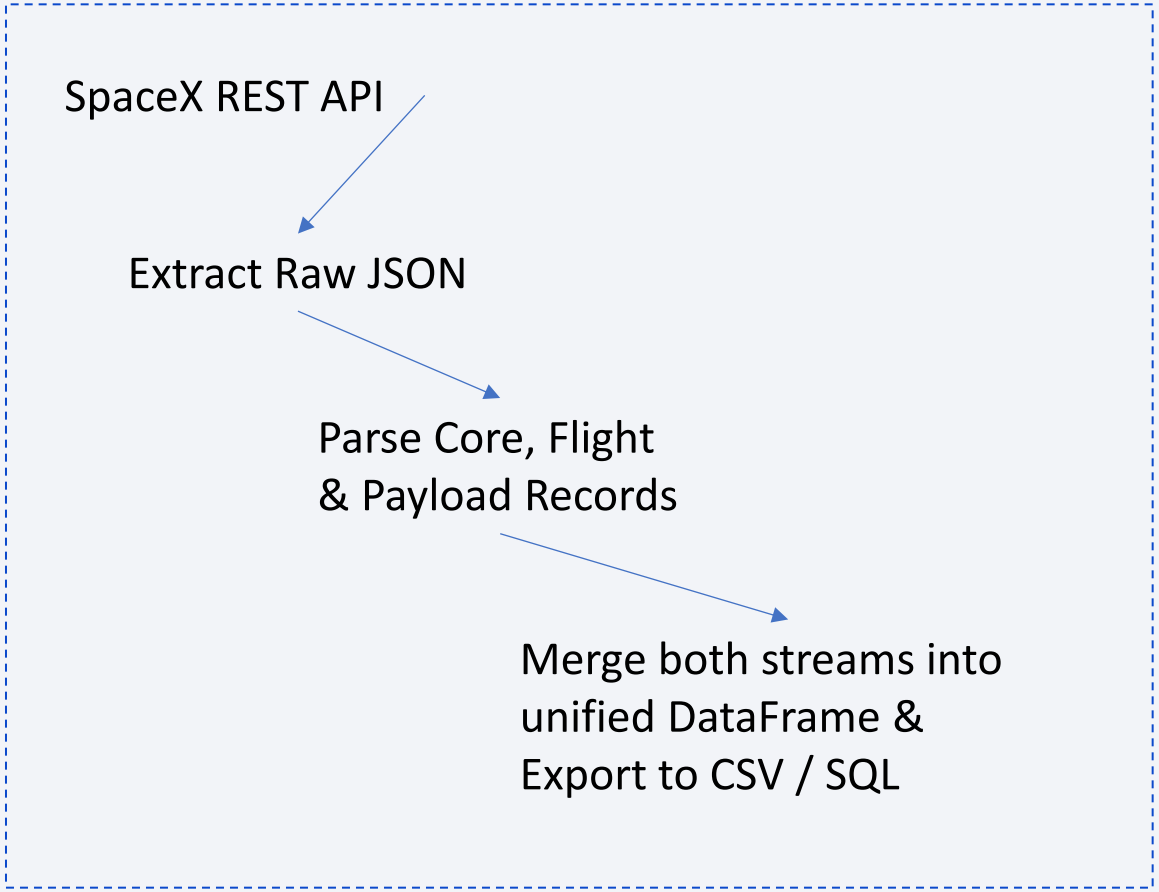
- Describe how data sets were collected.
  - Utilized the SpaceX REST API (v4/launches/past) to download structured JSON data containing core metrics, flight tracking numbers, payload masses, and landing outcomes.
- You need to present your data collection process use key phrases and flowcharts



Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts

- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose: https://github.com/hans1202/testrepo/blob/main/api.ipynb



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts

- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose

Wikipedia Pages → HTML Tables Scrape

↓

Standardize Elements& Mission Stats

→ Merge both streams into unified DataFrame & Export to CSV / SQL

Data Wrangling

- Describe how data were processed
- You need to present your data wrangling process using key phrases and flowcharts
- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose
  - https://github.com/hans1202/testrepo/blob/main/wrap.ipynb

EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
  - Scatter Plot, Bar Chart, Line Chart
- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose
  - https://github.com/hans1202/testrepo/blob/main/edadataviz.ipynb

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
  - Launch Site Filters
  - Calculated total payload mass
  - Found the first successful ground pad landing date using the MIN function
  - Counted total landing outcomes and used subqueries to identify boosters carrying maximum payload
- Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose
  - https://github.com/hans1202/testrepo/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

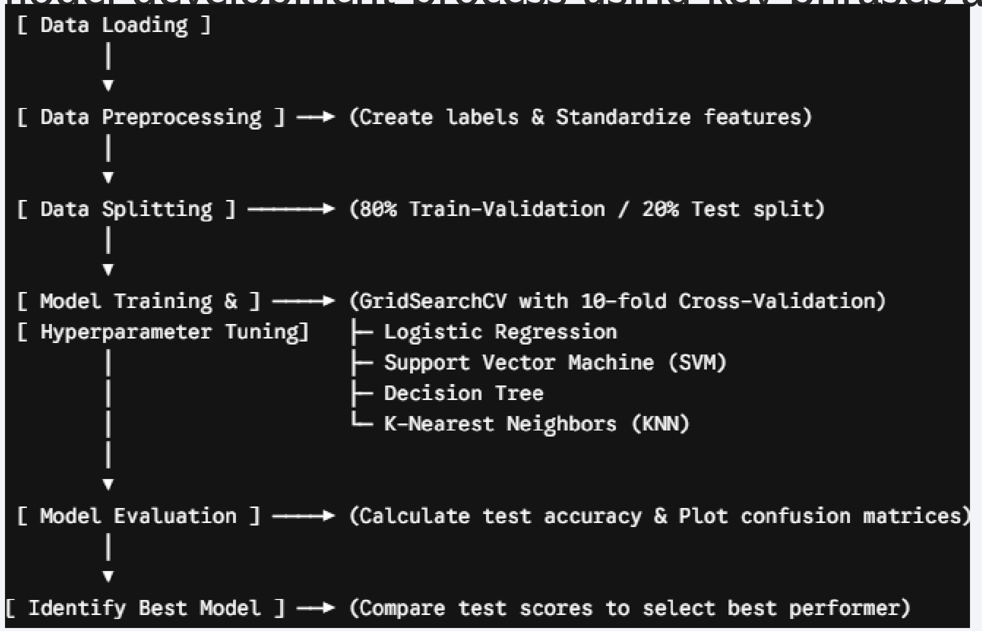
- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
  - folium.Circle, folium.Marker, DivIcon, folium.PolyLine
- Explain why you added those objects
  - **Circles** : Added around launch site coordinates to visually define the operational area and safety buffer zones.
  - **Color-Coded Markers** : Plotted launch markers (green for success, red for failure) clustered together to show historical performance.
  - **Custom Text Labels** : Plotted launch site names and displayed calculated distance measurements on the map.
  - **Polylines** : Drew straight lines from launch sites to the nearest coastlines, highways, and railways to evaluate logistical and safety proximity.
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose:https://github.com/hans1202/testrepo/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
  - Interactive Dropdown Menu (site-dropdown)
  - Interactive Range Slider (payload-slider)
  - Dynamic Pie Chart (success-pie-chart)
  - 4. Dynamic Scatter Plot with Color-Coded Categories (success-payload-scatter-chart)
- Explain why you added those plots and interactions
  - **Dropdown** and **Pie Chart** interaction allows stakeholders to analyze launch success， The **Scatter Plot** was added to reveal whether heavier payloads are more prone to failure， Integrating both the **Dropdown** and the **Range Slider** as concurrent inputs into the Scatter Plot allows the user to ask highly specific questions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose
  - https://github.com/hans1202/testrepo/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart



- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose
  - https://github.com/hans1202/testrepo/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

- Exploratory data analysis results
- **Total Samples**: The dataset contains 90 launch records in total (80% split into 72 training/validation samples and 20% into 18 test samples)
- **Target Class**: The binary target column Class indicates launch success (1 for a successful landing, 0 otherwise).
- **Imbalanced Classes**: The test set of 18 samples contains 15 successful landings (land) and 3 unsuccessful landings
- Interactive analytics demo in screenshots
- Predictive analysis results

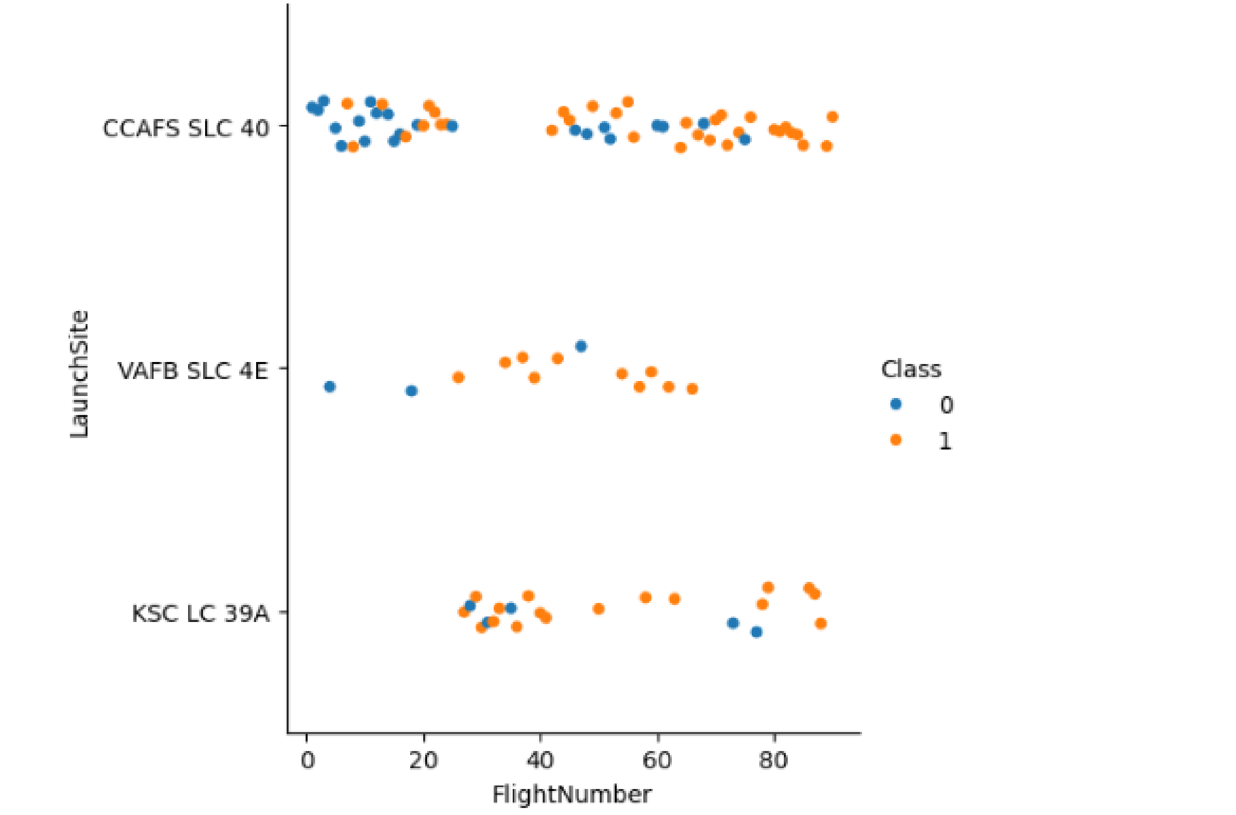
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site

- Show the screenshot of the scatter plot with explanations



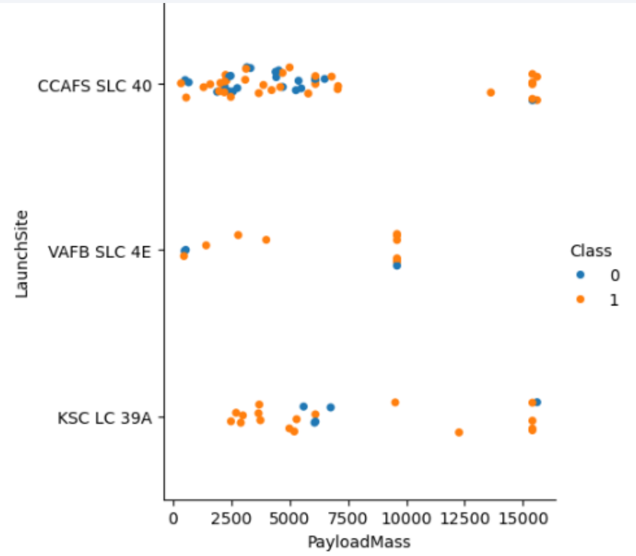
Technology Improved Over Time: Success rates (orange dots) increased significantly as flight numbers grew, showing a strong learning curve.

Site Reliability: KSC LC 39A achieved the highest launch success rate, while CCAFS SLC 40 hosted the earliest (and most unsuccessful) flights before stabilizing.

Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site

- Show the screenshot of the scatter plot with explanations

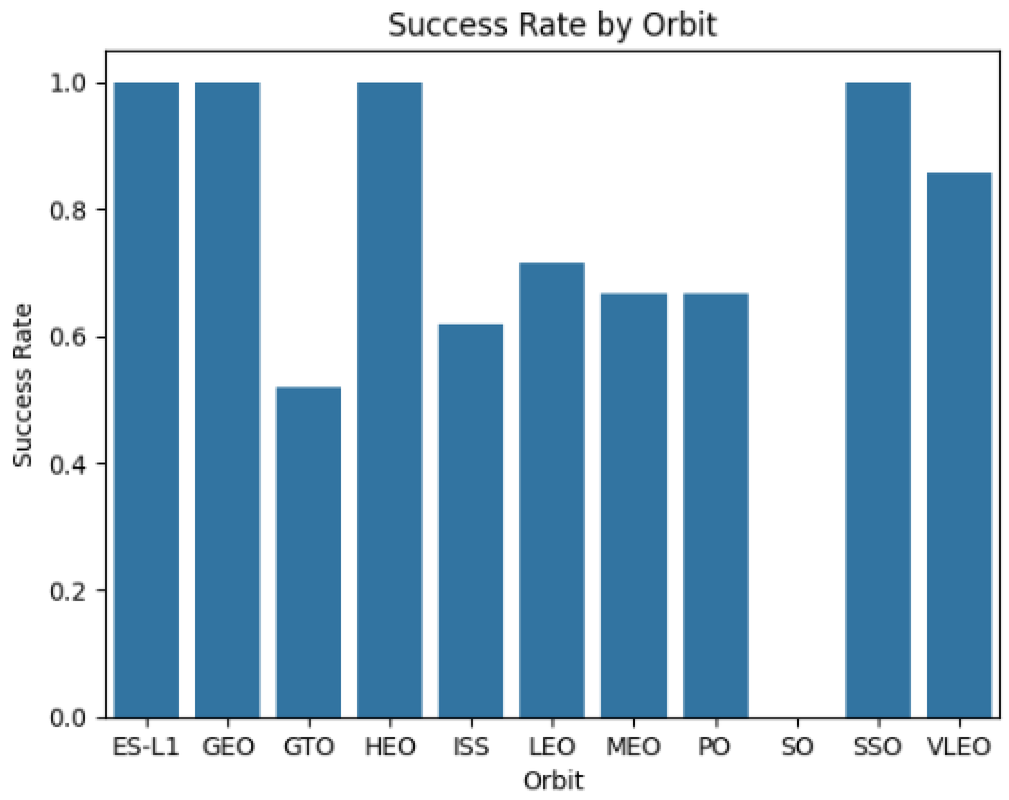


Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type

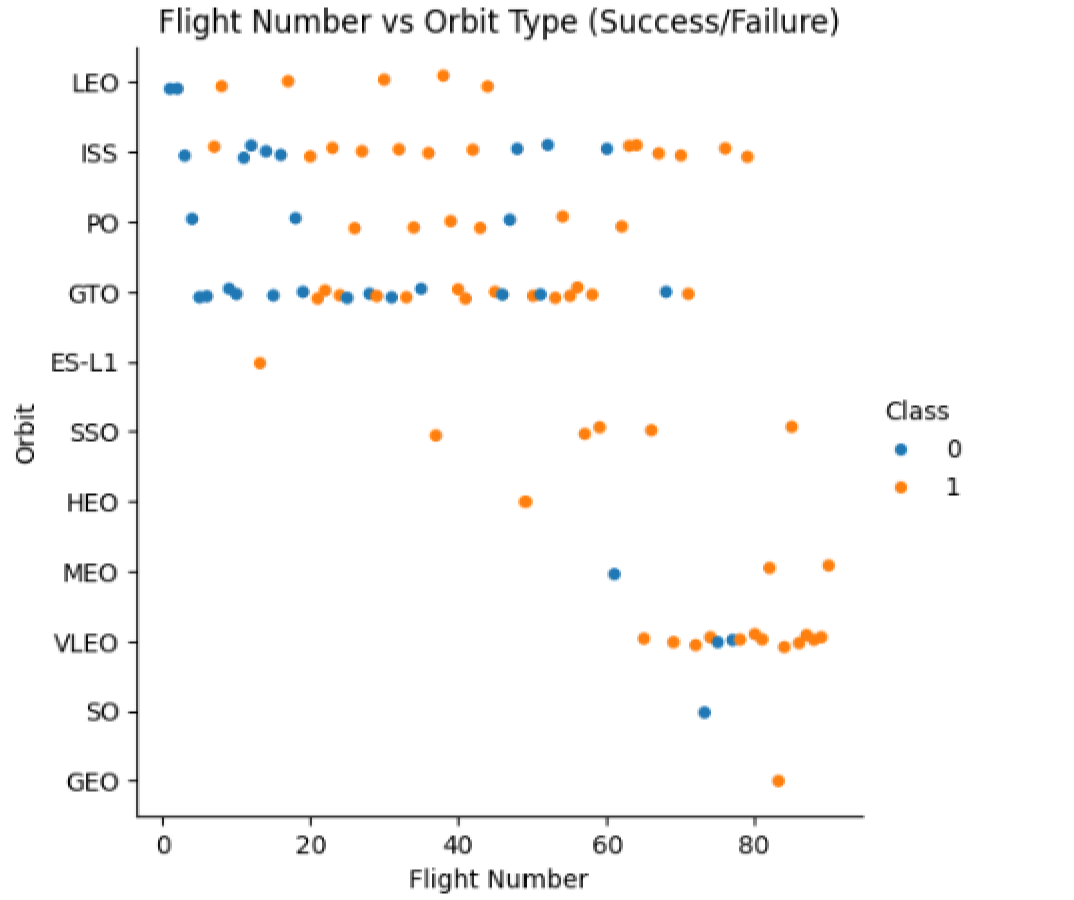
- Show the screenshot of the scatter plot with explanations




```
ES-LI,GEO,HEO AND SSO are most high
```

Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type
- Show the screenshot of the scatter plot with explanations

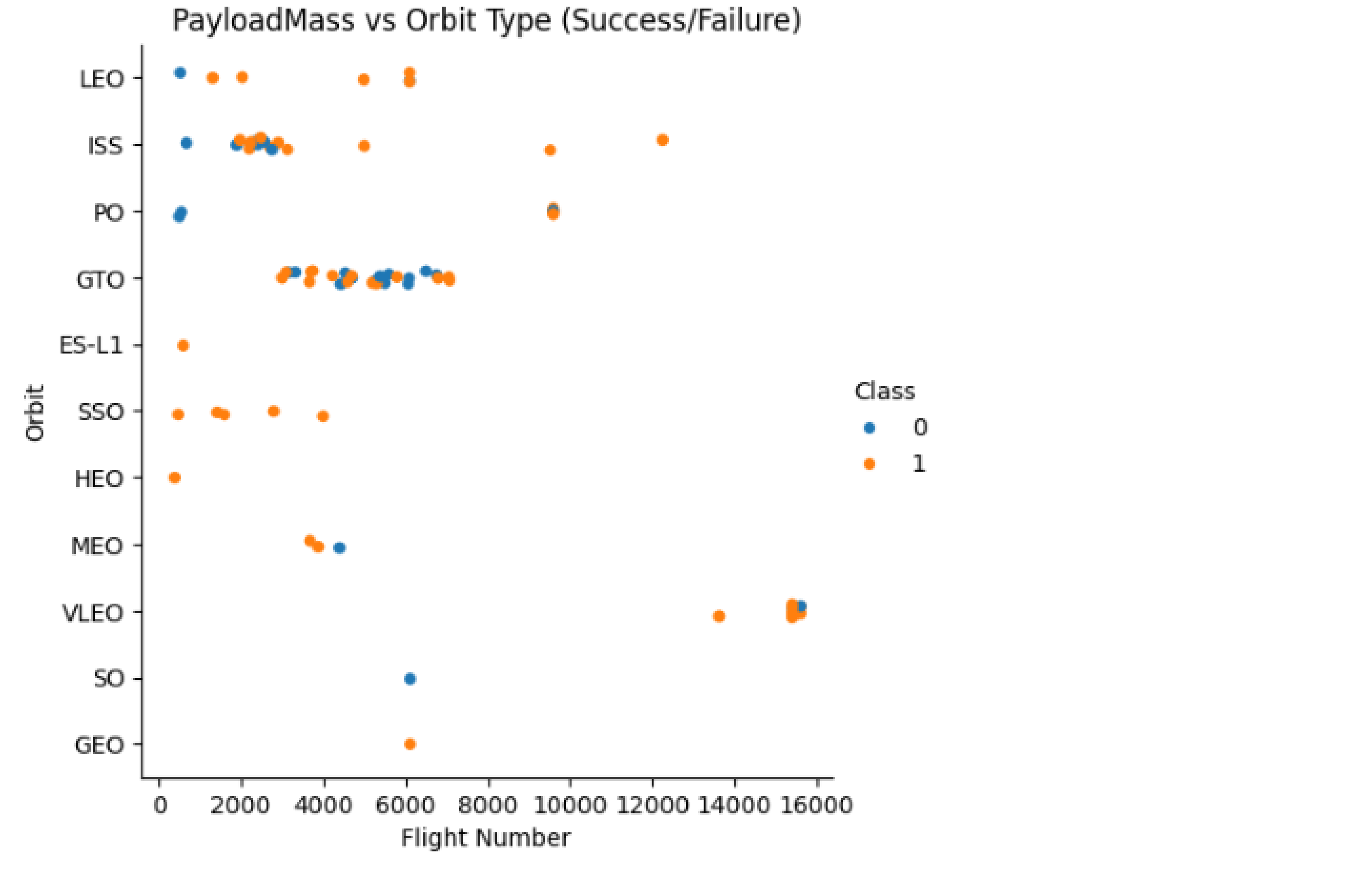


You can observe that in the LEO orbit, success seems to be related to the number of flights.

Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type

- Show the screenshot of the scatter plot with explanations



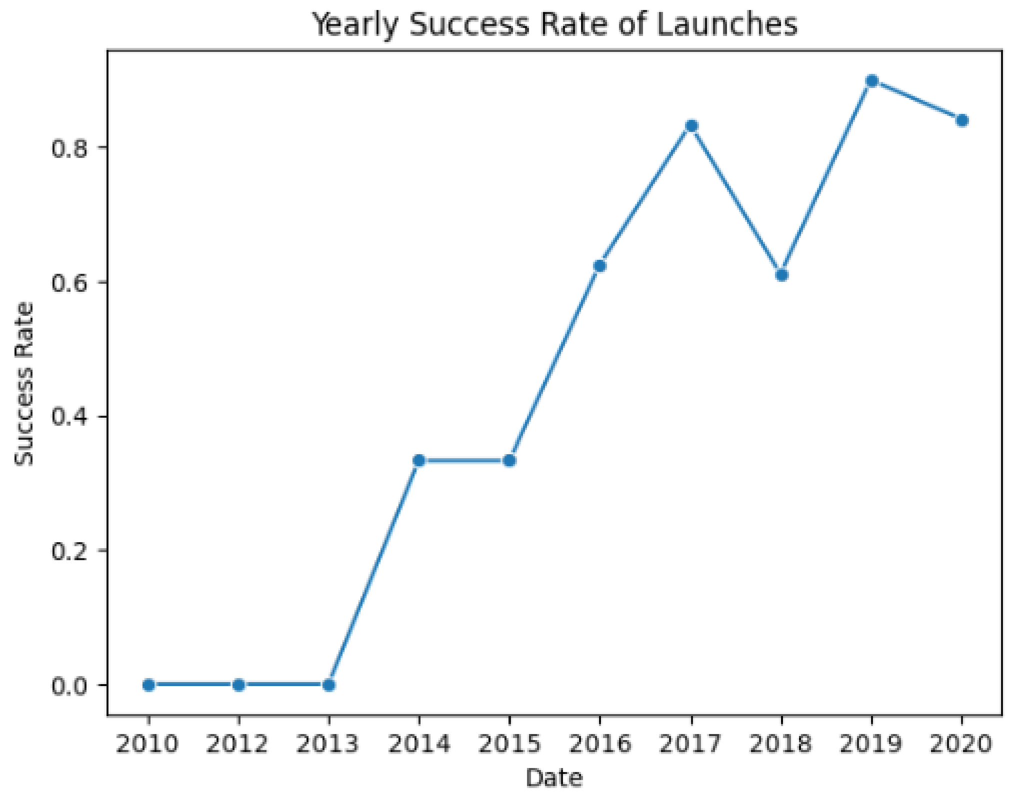
With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present

Launch Success Yearly Trend

- Show a line chart of yearly average success rate

- Show the screenshot of the scatter plot with explanations



you can observe that the sucess rate since 2013 kept increasing till 2020

All Launch Site Names

- Find the names of the unique launch sites
- Present your query result with a short explanation here

```
%%sql
SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- Present your query result with a short explanation here

```
%%sql
SELECT * FROM SPACEXTABLE WHERE "Launch_Site" like 'CCA%' limit 5 ;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight (
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- Present your query result with a short explanation here

```
%%sql
SELECT count(*) FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)' ;
```

```
 * sqlite:///my_data1.db
Done.
```

count(*)
20

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here

```
%%sql
SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1' ;
```

```
 * sqlite:///my_data1.db
Done.
```

AVG("PAYLOAD_MASS__KG_")
2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Present your query result with a short explanation here

```
%%sql
SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)' ;
```

```
 * sqlite:///my_data1.db
Done.
```

MIN("Date")
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

- Present your query result with a short explanation here

```
%%sql
SELECT Booster_Version FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND ("PAYLOAD_MASS__KG_"> 4000)  ;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1029.1
F9 FT B1021.2
F9 FT B1036.1
F9 B4 B1041.1
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Present your query result with a short explanation here

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here

```
%%sql
SELECT Booster_Version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Present your query result with a short explanation here

```
%%sql
SELECT
    substr(Date, 6, 2) AS Month,
    Mission_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTABLE
WHERE substr(Date, 1, 4) = '2015'
  AND Mission_Outcome like 'Failure%';
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Mission_Outcome	Booster_Version	Launch_Site
06	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%%sql
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

- Replace <Folium map screenshot 1> title with an appropriate title

- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map

- Explain the important elements and findings on the screenshot

<Folium Map Screenshot 2>

- Replace <Folium map screenshot 2> title with an appropriate title
- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map
- Explain the important elements and findings on the screenshot

<Folium Map Screenshot 3>

- Replace <Folium map screenshot 3> title with an appropriate title

- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed

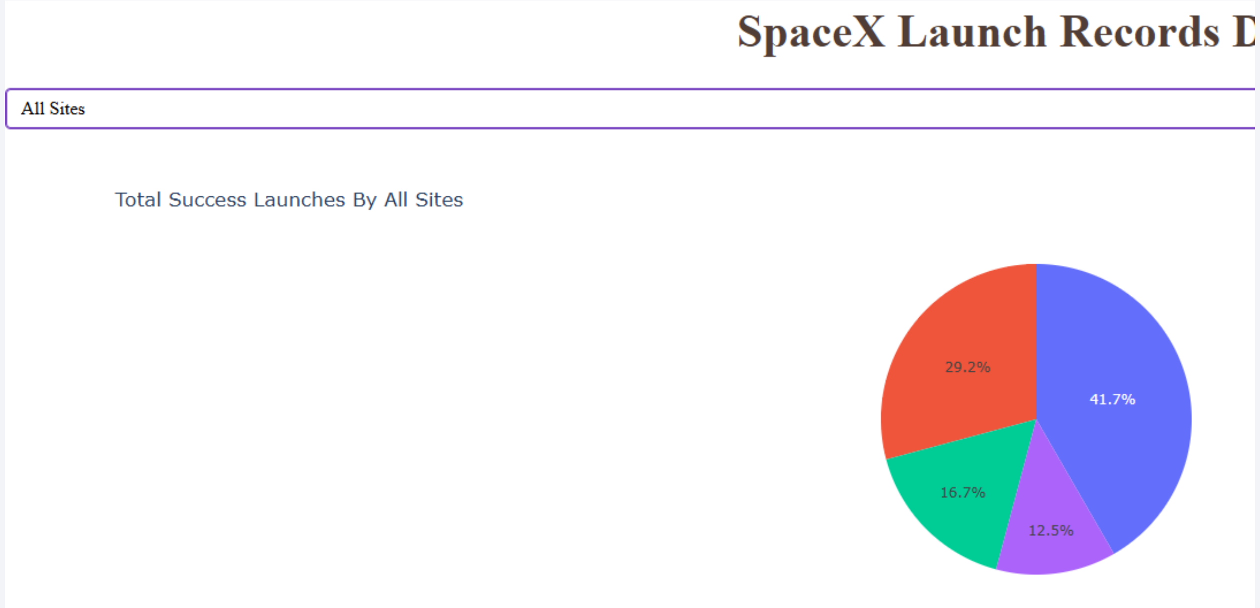
- Explain the important elements and findings on the screenshot

Section 4

Build a Dashboard with Plotly Dash

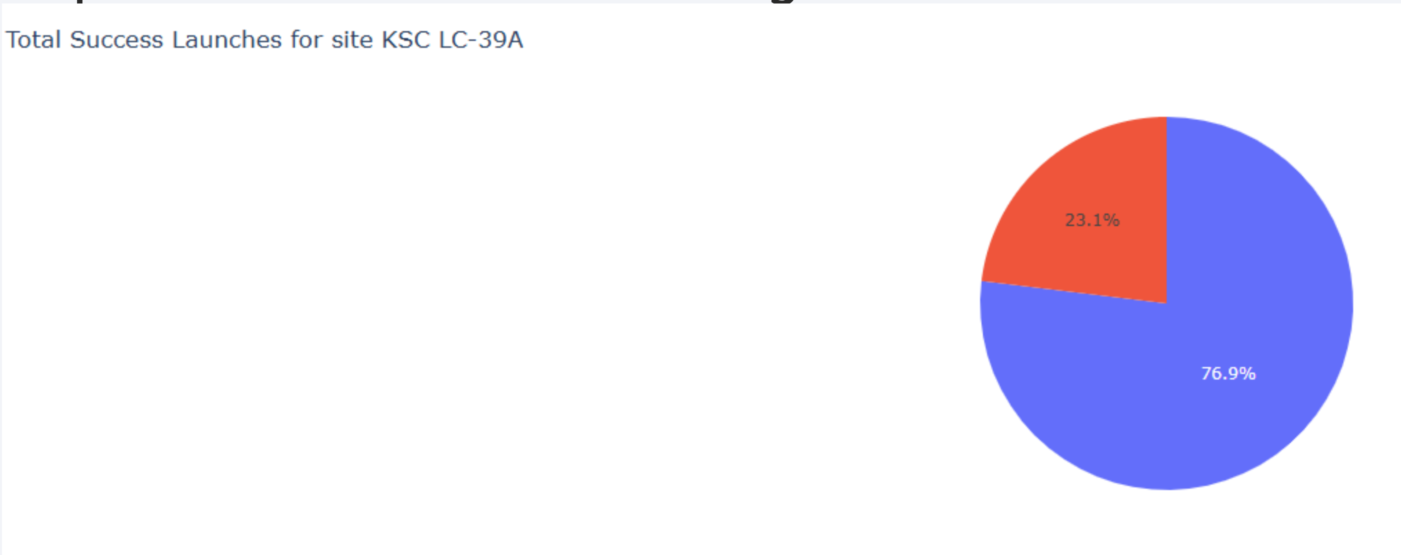
<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot



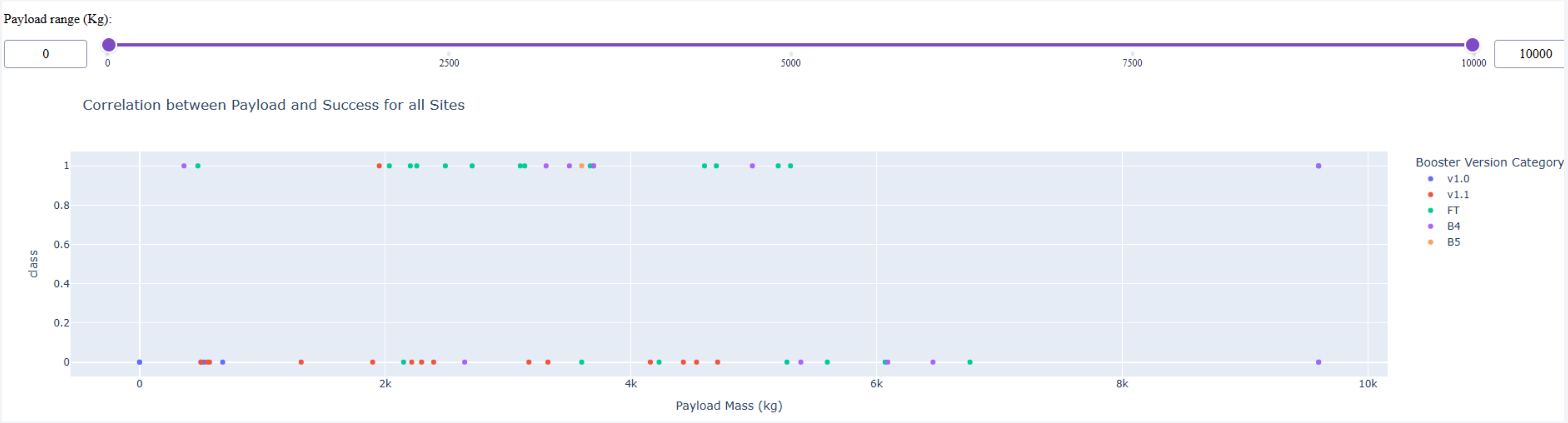
<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title
- Show the screenshot of the piechart for the launch site with highest launch success ratio
- Explain the important elements and findings on the screenshot



<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.



Section 5

Predictive Analysis (Classification)

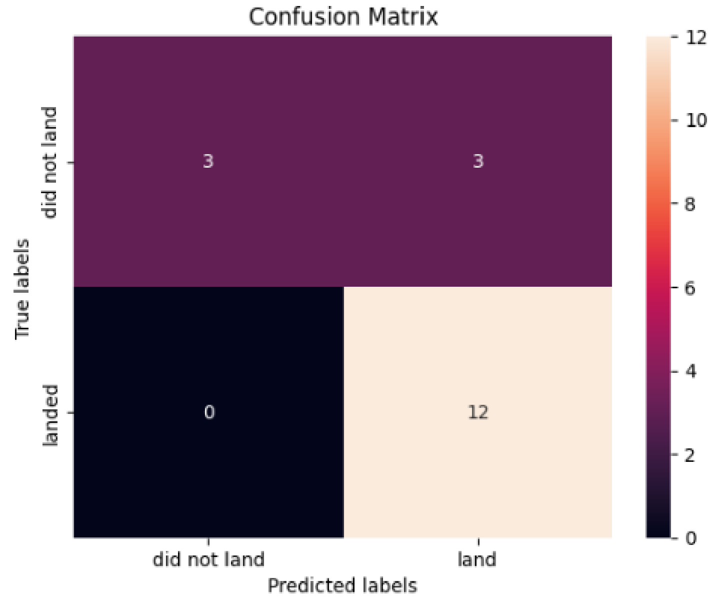
Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart

- Find which model has the highest classification accuracy
  - logistic regression

Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation



Examining the confusion matrix, we see that logistic regression can distinguish between the different classes. We see that the problem is false positives.

Overview:

True Postive - 12 (True label is landed, Predicted label is also landed)

False Postive - 3 (True label is not landed, Predicted label is landed)

Conclusions

- **High Accuracy on Landing Success:** The model correctly predicted all 12 successful landings (True Positives) with zero false negatives
- **issue with False Positives:** The main weakness is that all 3 failed landing attempts were incorrectly predicted as successful (False Positives), indicating the model struggle to identify failures due to class imbalance in the test set.
- …

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!